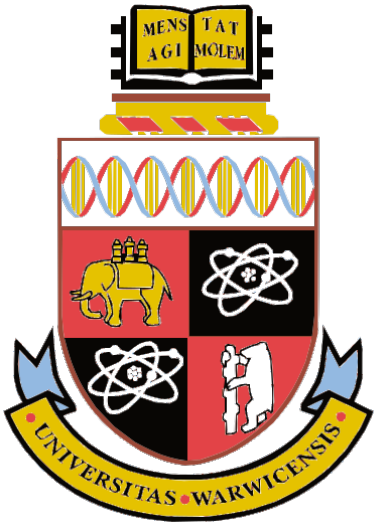


Quantum Speedup

by

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1 Introduction

This report examines what constitutes a quantum speedup. Understanding what constitutes a genuine quantum speedup requires a careful and intentional comparison of computational resources under matched model assumptions.

Throughout this report, the term *quantum speedup* is used to mean an asymptotic reduction in the computational resources needed to solve a specified problem when comparing the best quantum algorithm with the best classical algorithm, under the same model assumptions. Depending on the scenario, the appropriate resource might be circuit size (the number of elementary quantum gates) or, specifically in a black box setting, the number of oracle queries. This definition will be refined once we have built the necessary framework.

One of the central themes of this report is that quantum speedup is not tied to a single mechanism. Instead, it depends on both the computational model and the structure available in the problem.

The main case study is unstructured search, examining the role of Grover's algorithm in achieving a $\sqrt{N}$ speedup. As part of my contribution, I include Python simulations to illustrate the geometry and mechanisms of the algorithm. The full code can be seen in the appendices. I will also discuss phase estimation as a fundamentally different source of speedup since it uses spectral structure rather than amplitude amplification. I contrast amplitude amplification speedup with spectral structure speedup via phase estimation.

2 Preliminaries

This section fixes notation as well as elaborating on certain conventions used throughout. Proofs of key algebraic facts are deferred to section 3.

2.1 State spaces and the computational basis

In quantum computation the fundamental unit of computation is no longer the bit, but rather the quantum bit - the qubit. For $n \in \mathbb{N}$, the Hilbert space of n qubits is

$$\mathcal{H}_n := (\mathbb{C}^2)^{\otimes n} \cong \mathbb{C}^{2^n}.$$

When n is fixed and clear from context, simply write $\mathcal{H}$. Write $N := 2^n$ for the dimension of $\mathcal{H}_n$. [12, p. 13, 16–17]

The *state space* of a single qubit is $\mathbb{C}^2$, with computational (standard) basis vectors

$$|0\rangle := \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle := \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

An n-qubit classical outcome is represented by a bitstring $x = x_1x_2\cdots x_n \in \{0,1\}^n$. It tells us what the i-th qubit would read if you measured it in the computational basis.

We define

$$|x\rangle := |x_1\rangle \otimes \cdots \otimes |x_n\rangle,$$

so that $\{|x\rangle : x \in \{0,1\}^n\}$ is an orthonormal basis of $\mathcal{H}_n$, called the *computational basis.* [12, p. 13, 16–17]

Example 2.1 (States of a two-qubit system)**.** *For $n = 2$, we have*

$$|10\rangle = |1\rangle \otimes |0\rangle,$$

and there are four states in the computational basis

$$|00\rangle,\ |10\rangle,\ |01\rangle,\ |11\rangle.$$

If the system is in state $|01\rangle$, then measuring in the computational basis will return 0 for the first qubit and 1 for the second with probability 1.

The computational basis will be used to define the oracle in Grover's algorithm, as well as to interpret measurement outcomes as classical bitstrings.

2.2 Pure states and amplitudes

Definition 2.1 (Quantum state)**.** *A pure quantum state is a unit vector $|\psi\rangle \in \mathcal{H}_n$ satisfying $\langle\psi|\psi\rangle = 1$. Expanding a quantum state in the computational basis gives*

$$|\psi\rangle = \sum_{x\in\{0,1\}^n} \alpha_x|x\rangle, \qquad \alpha_x \in \mathbb{C}, \qquad \sum_x |\alpha_x|^2 = 1.$$

The coefficients α_x are called amplitudes*. [15, p. 115]*

Definition 2.2 (The Qubit)**.** *A qubit (quantum bit) is a pure quantum state in $\mathbb{C}^2$. [15, p. 74]*

Unlike a classical bit which is either 0 or 1, a qubit can be prepared in a superposition,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where the amplitudes α and β can vary continuously while satisfying $|\alpha|^2 + |\beta|^2 = 1$. However, measuring in the standard (computational) basis still produces an outcome of either 0 or 1, with probabilities $|\alpha|^2$ and $|\beta|^2$ respectively. These probabilities are given by the *Born rule.* [12, p. 85]

More generally, for an n-qubit state,

$$|\psi\rangle = \sum_x \alpha_x |x\rangle,$$

measurement in the computational basis returns a bitstring $x \in \{0,1\}^n$ with probability $|\alpha_x|^2$.

Remark 2.1. *For $\theta \in \mathbb{R}$, multiplying $|\psi\rangle$ by the phase $e^{i\theta}$ gives $e^{i\theta}|\psi\rangle$. This differs from $|\psi\rangle$ only by a* global phase. *This does not change any measurement properties so the vectors $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ represent the same physical state since they yield identical measurement statistics.*

This constraint– measurement reveals only one outcome x with probability $|\alpha_x|^2$, is what quantum speedups must exploit.

The Born rule is central here: since only a single sample from a larger probability distribution is seen, speedups can arise from reshaping the output distribution. This is exactly what Grover's algorithm does.

2.3 Bra–ket notation and inner products

A vector in $\mathcal{H}_n$ is written as a *ket* $|\psi\rangle$. Its conjugate transpose is the corresponding *bra*

$$\langle\psi| := (|\psi\rangle)^\dagger.$$

The inner product is written $\langle\phi|\psi\rangle \in \mathbb{C}$, and the outer product is $|\phi\rangle\langle\psi| \in \mathcal{L}(\mathcal{H}_n)$. The associated norm can be written

$$\| \, |\psi\rangle \, \| := \sqrt{\langle\psi|\psi\rangle}.$$

An orthonormal basis satisfies

$$\langle x|y\rangle = \delta_{x,y} \qquad \text{for all basis vectors } |x\rangle \text{ and } |y\rangle.$$

This notation is convenient because it distinguishes state vectors $|\psi\rangle$ from dual vectors $\langle\phi|$, and makes the use of inner products $\langle\phi|\psi\rangle$ and outer products $|\phi\rangle\langle\psi| \in \mathcal{L}(\mathcal{H}_n)$ more compact. [12, p. 66–68]

2.4 Projective measurement

Definition 2.3 (Projective measurement)**.** *A projective measurement is specified by a family of orthogonal projectors $\{P_i\}_{i\in I}$ on $\mathcal{H}$ satisfying*

$$P_i P_j = \delta_{ij} P_i, \qquad \sum_{i\in I} P_i = I.$$

If the system is in state $|\psi\rangle$, then outcome i occurs with probability

$$\mathbb{P}(i) = \langle\psi|P_i|\psi\rangle,$$

and given that outcome i occurs, the post-measurement state is

$$\frac{P_i|\psi\rangle}{\|P_i|\psi\rangle\|} \qquad \textit{(provided } \mathbb{P}(i) > 0\textit{)}.$$

Measuring in the computational basis corresponds to projectors

$$P_x = |x\rangle\langle x| \qquad \textit{for } x \in \{0,1\}^n.$$

[12, p. 87–90]

Example 2.2 (1-qubit measurement)**.** *Let*

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

Measuring in the computational basis gives projectors,

$$P_0 = |0\rangle\langle 0|, \qquad P_1 = |1\rangle\langle 1|.$$

The probabilities of the outcomes are

$$\mathbb{P}(0) = \langle\psi|P_0|\psi\rangle = |\alpha|^2, \qquad \mathbb{P}(1) = \langle\psi|P_1|\psi\rangle = |\beta|^2.$$

If 0 occurs, then the post-measurement state,

$$\frac{P_0|\psi\rangle}{\sqrt{\mathbb{P}(0)}} = |0\rangle$$

and similarly with the occurrence of 1.

Example 2.3 (two-qubit measurement)**.** *Let*

$$|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

Measurement of $|\psi\rangle$ in the computational basis returns each of the outcomes

$$|00\rangle,\ |01\rangle,\ |10\rangle,\ |11\rangle$$

with equal probability 1/4.

In particular, measurement returns the outcome 10 *with probability* 1/4*. If the measurement outcome is* 10*, then the post-measurement state updates to* $|10\rangle$*.*

A common misconception is that quantum algorithms "read all outcomes at once". This is not true. Example 2.3 shows that creating a superposition does not reveal all values of a function; measurement returns a single sample from a larger probability distribution. The advantage of a quantum algorithm lies in its ability to increase the probability of a desired outcome before measurement, as in amplitude amplification.

Remark 2.2 (Repeatability of projective measurement)**.** *If a projective measurement $\{P_i\}$ is performed and outcome i occurs, then the post-measurement state is proportional to $P_i|\psi\rangle$ (the state points in the direction $P_i|\psi\rangle$).*

In particular, if $|\psi\rangle$ is such that $P_i|\psi\rangle = |\psi\rangle$, then outcome i occurs with probability 1 *and the state remains unchanged. Consequently, if the same measurement is repeated, it will return the same outcome with probability* 1*. [7, p. 2–4]; [12, p. 87–90]*

2.5 Unitaries

A linear operator $U : \mathcal{H} \to \mathcal{H}$ is *unitary* if

$$U^\dagger U = UU^\dagger = I.$$

Unitary operators preserve inner products and norms, hence map quantum states to quantum states. In Section 3, the query and circuit model are formalised, and properties of unitary operators are used to justify certain algorithmic steps.

3 Mathematical framework of quantum computation

This section is a bridge between the preliminaries and the later analysis. It introduces the circuit and query models used throughout this report.

3.1 Quantum circuits

Definition 3.1 (Quantum gate)**.** *A quantum gate acting on n qubits is specified by a unitary operator*

$$U : \mathcal{H}_n \to \mathcal{H}_n,$$

and it is represented by a $2^n \times 2^n$ unitary matrix in the computational basis. [9, p. 61]; [12, p. 18–20]

In practice, most gates act on only $k \ll n$ qubits; these gates can viewed as operators on $\mathcal{H}_n$ by tensoring with the identity on the remaining qubits, i.e.

$$U \mapsto U \otimes I_{2^{n-k}},$$

up to the ordering of the qubits.

In classical computation, circuits are built from logic gates. Quantum gates play the same structural role. If a circuit applies gates $U_1, \ldots, U_m$ in order, then the overall unitary operation is

$$U = U_m \cdots U_1.$$

Definition 3.2 (Quantum circuit)**.** *A quantum circuit is a sequence of quantum gates acting on a collection of qubits.*

The size *of a circuit refers to the number of gates it contains. The* depth *of a circuit refers to the minimum number of time steps needed when the gates acting on disjoint sets of qubits are applied in parallel. [9, p. 61]*

Example 3.1. *Consider a two-qubit circuit that first applies a H gate to the first qubit and then a* CNOT *gate to both qubits. Then the overall unitary of the circuit is*

$$U = \mathrm{CNOT}(H \otimes I).$$

This illustrates that successive gates compose by matrix multiplication.

The circuit model allows a quantum computation to be written as a sequence of elementary unitary operations acting on qubits.

Unlike classical logic gates, which do not need to be reversible, quantum gates are necessarily reversible due to being unitaries. This is an important distinction used in the design of quantum algorithms like Grover's algorithm.

Black-box access will be introduced later, where querying the input is treated as a special operation and counted separately from the other gates in the circuit.

3.2 Unitary evolution

Definition 3.3 (Unitary evolution)**.** *Before measurement, the state of an n-qubit quantum system can change according to unitary operators. If the system is in state $|\psi\rangle$ and a circuit implementing the unitary U is applied, the resulting state is*

$$|\psi\rangle \mapsto U|\psi\rangle.$$

[12, p. 84]

Unitary maps preserve inner products and hence preserve total probability, so a unitary evolution maps a quantum state to a quantum state.

3.3 Measurement as the output rule

As discussed in Section 2, measurement produces a classical outcome from the probability distribution determined by the amplitudes of the state. After preparing the qubits and

applying the gates, the final state will be described by

$$|\psi\rangle = \sum_x \alpha_x |x\rangle,$$

but one cannot directly read off the amplitudes α_x. To get an actual output, a measurement must be performed. Measuring in the computational basis will return outputs $x \in \{0,1\}^n$ with probability $|\alpha_x|^2$. After the measurement, the state collapses to $|x\rangle$. Call this the post-measurement state.

Therefore, before measurement, a quantum algorithm can influence the outcome by shaping this probability distribution.

3.4 Black-box problems

Definition 3.4 (Oracle and query)**.** *Let $f : \{0,1\}^n \to \{0,1\}^m$ be a function. An* oracle *for f is an operation that gives access to the values of f without needing an explicit description of how f is computed. Oracle access is modelled by a unitary operator O_f.*

A query *is a single application of the oracle, that is, one use of O_f. [9, p. 95]*

Definition 3.5 (Black-box/oracle problem)**.** *A black-box problem is a computational task in which no input is given explicitly, but the input can only be accessed through queries to an oracle.*

Typically, the input is an unknown function f (from a specified family, like $f : \{0,1\}^n \to \{0,1\}$) and an algorithm gains information about f only via queries to an oracle. [9, p. 95]

3.5 The quantum query model

Definition 3.6 (Quantum query/black-box model)**.** *Let $f : \{0,1\}^n \to \{0,1\}$ be an unknown Boolean function, accessed only through an oracle unitary O_f.*

*A T-*query quantum algorithm *starts with an initial state $|\psi_{in}\rangle$ and uses a sequence of input-independent unitaries $U_0, U_1, \ldots, U_T$. The final state the algorithm produces has the form*

$$|\psi_{out}\rangle \;=\; U_T\, O_f\, U_{T-1}\, O_f \cdots O_f\, U_1\, O_f\, U_0\, |\psi_{in}\rangle.$$

The algorithm outputs a classical value by performing a measurement on $|\psi_{out}\rangle$. The (quantum) query complexity of the algorithm is the number of oracle calls T. [1, p. 222]

3.6 Cost models: complexity

When discussing quantum speedup, one must compare a quantum algorithm to the best classical algorithm. Different models measure different resources, so the comparison must have the cost measure fixed in advance to be meaningful.

In this report, two standard models are used. In the *circuit (gate) model*, the cost of an algorithm is measured by the number of elementary quantum gates (the circuit size); one could also use circuit depth and the number of qubits used. For black-box problems, it is more conventional to use the *quantum query model*, in which the input is accessed through an oracle O_f and the cost is the number of oracle calls (queries).

3.7 Unstructured search

Example 3.2 (Unstructured search)**.** *Let $N = 2^n$ and let $f : \{0,1\}^n \to \{0,1\}$ be a Boolean function which satisfies*

$$\sum_{x \in \{0,1\}^n} f(x) = 1,$$

so there exists a unique marked input x^* with $f(x^*) = 1$. Given oracle access to f, the task is to output x^* with high probability. The cost of an algorithm in the query model is the number of oracle calls.

In classical computing, this can be visualised as N boxes. To find the marked box x^*, one must inspect each box one at a time since no structural information is given about f. Classical queries are expected to be $N/2$ for randomised search, with N being the worst case scenario.

3.8 Phase oracle for unstructured search

In Grover search, one is given a Boolean function

$$f : \{0,1\}^n \to \{0,1\}$$

where $f(x^*) = 1$ if and only if x^* is marked. In the simplest case, the unique marked item x^* is such that

$$f(x^*) = 1, \qquad f(x) = 0 \text{ for } x \neq x^*.$$

The case of multiple marked states is a natural extension of what is discussed in Grover's algorithm; the same two-dimensional geometry applies.

The *phase oracle* is the unitary O_f defined on computational basis states by

$$O_f|x\rangle = (-1)^{f(x)}|x\rangle.$$

Equivalently, O_f flips the phase of marked basis states and acts as the identity on unmarked basis states.

3.9 Oracle as a unitary operator

Lemma 3.1 (The phase oracle is unitary)**.** *Let $f : \{0,1\}^n \to \{0,1\}$ and define the phase oracle by*

$$O_f|x\rangle = (-1)^{f(x)}|x\rangle.$$

Then O_f is unitary. In particular, $O_f^\dagger = O_f$ and $O_f^2 = I$. [12, 249]

Proof. Since $\{|x\rangle\}$ is an orthonormal basis, it suffices to determine $O_f^\dagger$ and O_f^2 on basis states. For any x, $(-1)^{f(x)} \in \{-1,1\} \subset \mathbb{R}$, hence

$$O_f^\dagger|x\rangle = \overline{(-1)^{f(x)}}\,|x\rangle = (-1)^{f(x)}|x\rangle = O_f|x\rangle,$$

so $O_f^\dagger = O_f$. Moreover,

$$O_f^2|x\rangle = O_f\big((-1)^{f(x)}|x\rangle\big) = (-1)^{f(x)}(-1)^{f(x)}|x\rangle = (-1)^{2f(x)}|x\rangle = |x\rangle,$$

so $O_f^2 = I$. Therefore $O_f^\dagger O_f = O_f^2 = I$, and O_f is unitary. □

The oracle can then be viewed as the diagonal unitary [3, p. 1]

$$O_f = \sum_{x\in\{0,1\}^n} (-1)^{f(x)}\,|x\rangle\langle x| = \begin{pmatrix} (-1)^{f(0\ldots0)} & 0 & \cdots & 0 \\ 0 & (-1)^{f(0\ldots1)} & \ddots & 0 \\ \vdots & \ddots & \ddots & 0 \\ 0 & 0 & 0 & (-1)^{f(1\ldots1)} \end{pmatrix}$$

The framework developed in this section sets the stage for Grover's algorithm to be analysed. The query model gives us a notion of oracle cost and the phase oracle is the mechanism by which marked states are distinguished. The next sections use this framework to derive Grovers algorithm.

4 Where speedup comes from: interference, geometry and measurement limits

In this section, the three mechanisms used in Grover's quantum speedup are explained. First, linearity allows amplitudes to interfere. Second, the oracle and diffusion operators act geometrically as reflections, whose composition becomes a rotation in a two-dimensional subspace. Third, measurement reveals only a single outcome, so the usefulness of a quantum state is dependent on how much probability mass is concentrated on a desirable outcome.

4.1 Linearity and interference

Linearity allows amplitudes to interfere before measurement. If U is linear and

$$|\psi\rangle = \sum_x \alpha_x |x\rangle$$

then,

$$U|\psi\rangle = \sum_x \alpha_x U|x\rangle.$$

Hence a single basis state in $U|\phi\rangle$ might be built from several contributions, meaning several terms may interfere constructively or destructively.

Example 4.1 (The Hadamard gate action)**.** *The Hadamard gate acts on the computational basis by*

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Starting with the initial state $|\psi_0\rangle = |0\rangle$, one application of H gives,

$$|\psi_1\rangle \ = H|0\rangle \ = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Applying H again gives

$$|\psi_2\rangle = \frac{1}{\sqrt{2}}\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} + \frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) = \frac{1}{2}(|0\rangle + |1\rangle + |0\rangle - |1\rangle) = \frac{1}{2}(2|0\rangle) = |0\rangle$$

In this example, the amplitudes of $|1\rangle$ cancel and the $|0\rangle$ add; this is destructive and constructive interference respectively. Quantum algorithms are designed to interfere destructively for undesirable outcomes, and constructively for desirable outcomes.

4.2 Geometry of unitaries: reflections and rotations

Lemma 4.1 (Reflection is unitary)**.** *Let $|u\rangle$ be a unit vector, and define*

$$R_u := I - 2|u\rangle\langle u|.$$

Then R_u is unitary. Moreover, R_u acts as a reflection in the hyperplane orthogonal to $|u\rangle$.

Proof. A straight forward calculation shows this. Note that $(|u\rangle\langle u|)^\dagger = (\langle u|)^\dagger(|u\rangle)^\dagger = |u\rangle\langle u|$, so

$$R_u^\dagger = I - 2|u\rangle\langle u| = R_u.$$

Also, since $\langle u \mid u\rangle = 1$,

$$(|u\rangle\langle u|)^2 = |u\rangle\langle u|u\rangle\langle u| = \langle u \mid u\rangle|u\rangle\langle u| = |u\rangle\langle u|.$$

Then,

$$R_u^\dagger R_u = R_u^2 = (I - 2|u\rangle\langle u|)^2 = I - 4(|u\rangle\langle u|) + 4(|u\rangle\langle u|)^2 = I.$$

Hence $R_u^\dagger R_u = I$, so R_u is a unitary.

Now let $|\psi\rangle = \alpha|u\rangle + |v\rangle$, where $\langle u \mid v\rangle = 0$. Then

$$R_u|\psi\rangle = (I - 2|u\rangle\langle u|)(\alpha|u\rangle + |v\rangle) = -\alpha|u\rangle + |v\rangle.$$

Thus R_u reverses the component parallel to $|u\rangle$ and the orthogonal component remains unchanged. This is exactly reflection in the hyperplane $|u\rangle^\perp$. □

Remark 4.1. *Note that the reflection R_u is both unitary and Hermitian, implying it is also a involution ($R_u^2 = I$).*

Theorem 4.1 (Product of reflections is a rotation)**.** *Let $|u\rangle$ and $|v\rangle$ be linearly independent unit vectors, and define the corresponding reflections as*

$$R_u := I - 2|u\rangle\langle u|, \qquad R_v := I - 2|v\rangle\langle v|.$$

Let the angle between the vectors be

$$\theta := \arccos|\langle u|v\rangle| \in (0, \pi/2].$$

Then on the two-dimensional subspace

$$\mathrm{span}\{|u\rangle, |v\rangle\},$$

the operator $R_u R_v$ is a rotation by angle 2θ. Moreover,

$$\mathrm{span}\{|u\rangle, |v\rangle\}^\perp$$

is invariant under both reflections, and hence is invariant under $R_u R_v$.

Proof sketch. On the orthogonal complement of $\mathrm{span}\{|u\rangle, |v\rangle\}$, each reflection acts as the identity, so it is enough to restrict attention to this two-dimensional subspace. After choosing an orthonormal basis $\{|u\rangle, |u_\perp\rangle\}$ for it, one finds that $R_u R_v$ has a matrix of the form

$$\begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}.$$

This is exactly the matrix of a rotation by angle 2θ. Due to the length, a full proof is given in Appendix A. [12, p. 252] □

Therefore, when an algorithm is built from two reflections, their composition acts as a

rotation by twice the angle between them on a two-dimensional subspace. This is the key geometric idea behind Grover's algorithm.

4.3 Limits of measurement

Although an n-qubit pure state lies in an $N = 2^n$ dimensional Hilbert space, the output of a quantum measurement is a classical– it is limited. Measurement does not reveal the all the amplitudes. Instead, given a state

$$|\psi\rangle = \sum_x \alpha_x |x\rangle,$$

by the Born rule, the measurement returns outcome x with probability

$$|\alpha_x|^2$$

therefore usefulness of a quantum state is not determined by the number of basis states which appear in the superposition, but by how much probability mass is concentrated on desirable outcomes. This is why amplitude amplification is useful in algorithms.

Lemma 4.2 (Amplification is only useful via measurement)**.** *Let $M \subseteq \{0,1\}^n$ be the set of outcomes, and let*

$$\Pi_M := \sum_{t \in M} |t\rangle\langle t|$$

be the projector onto the span of $\{|t\rangle : t \in M\}$. If

$$|\psi\rangle = \sum_x \alpha_x |x\rangle$$

is the pre-measurement state, then the probability that a measurement in the computational basis outputs an element of M is

$$\mathbb{P}[\textit{output} \in M] = \langle\psi|\Pi_M|\psi\rangle = \sum_{t \in M} |\alpha_t|^2.$$

[5, p. 4]

Proof. By Definition 2.3,

$$\mathbb{P}[\text{output} \in M] = \langle\psi|\Pi_M|\psi\rangle.$$

It remains to show that

$$\langle\psi|\Pi_M|\psi\rangle = \sum_{t \in M} |\alpha_t|^2.$$

First, by direct calculation,

$$\begin{aligned}\Pi_M|\psi\rangle &= \left(\sum_{t\in M}|t\rangle\langle t|\right)\left(\sum_x \alpha_x|x\rangle\right)\\ &= \sum_{t\in M}\sum_x \alpha_x|t\rangle\langle t|x\rangle\\ &= \sum_{t\in M}\sum_x \alpha_x|t\rangle\,\delta_{t,x}\\ &= \sum_{t\in M}\alpha_t|t\rangle.\end{aligned}$$

Therefore,

$$\begin{aligned}\langle\psi|\Pi_M|\psi\rangle &= \left(\sum_y \overline{\alpha_y}\langle y|\right)\left(\sum_{t\in M}\alpha_t|t\rangle\right)\\ &= \sum_{t\in M}\sum_y \overline{\alpha_y}\alpha_t\langle y|t\rangle\\ &= \sum_{t\in M}\sum_y \overline{\alpha_y}\alpha_t\,\delta_{y,t}\\ &= \sum_{t\in M}\overline{\alpha_t}\alpha_t\\ &= \sum_{t\in M}|\alpha_t|^2.\end{aligned}$$

□

This lemma shows us the limitation and the misconception of quantum parallelism. Although linearity allows a unitary to act on all basis components simultaneously, measurement limits us so that it only reveals one of those outcomes. As a result, any quantum speedup in a black-box setting must be made by reshaping the distributions in someway to increase the probability of certain outcomes.

The next section demonstrates how Grover's algorithm exploits this reshaping by composing two reflections to produce a rotation and how this results in concentrated amplitude on a marked state.

5 Main case study: Grover's algorithm

This section formulates the search problem, builds the Grover iterate from two reflections, derives the success probability after k iterations, then explains why this gives the optimal $\mathcal{O}(\sqrt{N})$ query complexity in the black box model.

5.1 Problem setup

Consider the unstructured search problem in example 3.2, where $N = 2^n$ and

$$f : \{0,1\}^n \to \{0,1\}$$

be a Boolean function with a unique marked input x^*, such that

$$f(x^*) = 1, \quad f(x) = 0 \text{ for all } x \neq x^*.$$

Given oracle access to f, the task is to output x^* with high probability.

In the quantum query model (Section 3), access is provided through the phase oracle

$$O_f|x\rangle = (-1)^{f(x)}|x\rangle.$$

which is unitary by Lemma 3.1. One application of O_f (an oracle call) counts as a single query, and the total number of oracle calls is the query complexity.

Grover's algorithm starts with the superposition state,

$$|\phi\rangle := \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} |x\rangle.$$

By the Born rule (Section 2.4), and the projection calculation in Lemma 4.2, the probability of measuring the marked state is the squared magnitude of its amplitude in the computational basis. The aim is to apply unitaries that increase the amplitude of $|x^*\rangle$ before measurement.

Remark 5.1. *Grover's algorithm extends naturally to the case of multiple marked items. For simplicity, however, this section is restricted to the case with one unique marked item. In the case of a unique marked item,*

$$O_f = I - 2|x^*\rangle\langle x^*|,$$

since the oracle flips the phase of $|x^*\rangle$, and acts trivially on all the basis states orthogonal to it.

5.2 Grover's algorithm and constructing the Grover iterate

Up to global phase, Grover's iterate is a product of two reflections. Recall a reflection unitary has the form $R_u := I - 2|u\rangle\langle u|$ (Lemma 4.2). This transformation sends $|u\rangle \to -|u\rangle$ and fixes every vector orthogonal to $|u\rangle$.

Phase shift on the marked state: In the unique marked item case, the phase oracle flips the sign of the marked basis state while leaving every basis state orthogonal to it

unchanged. Thus

$$O_f|x^*\rangle = -|x^*\rangle, \qquad O_f|x\rangle = |x\rangle \quad \text{for } x \neq x^*.$$

Therefore it is clear that the oracle is exactly the reflection

$$O_f = I - 2|x^*\rangle\langle x^*|.$$

[5, p. 54]

Diffusion operator: Define the diffusion operator (*inversion about the mean*) by

$$D := 2|\phi\rangle\langle\phi| - I.$$

[5, p. 54]

Let

$$|\psi\rangle = \sum_x \alpha_x|x\rangle, \qquad |\phi\rangle = \frac{1}{\sqrt{N}}\sum_x |x\rangle,$$

and the mean amplitude is

$$\overline{\alpha} := \frac{1}{N}\sum_x \alpha_x.$$

Since

$$\langle\phi|\psi\rangle = \frac{1}{\sqrt{N}}\sum_x \alpha_x = \sqrt{N}\,\overline{\alpha},$$

it follows that

$$\begin{aligned} D|\psi\rangle &= (2|\phi\rangle\langle\phi| - I)|\psi\rangle \\ &= 2|\phi\rangle\langle\phi|\psi\rangle - |\psi\rangle \\ &= 2\left(\frac{1}{\sqrt{N}}\sum_x |x\rangle\right)\left(\frac{1}{\sqrt{N}}\sum_y \alpha_y\right) - \sum_x \alpha_x|x\rangle \\ &= \sum_x (2\overline{\alpha} - \alpha_x)|x\rangle. \end{aligned}$$

So, an amplitude α_x is reflected to $2\overline{\alpha} - \alpha_x$. It is reflected about the mean amplitude. [9, p. 158]

Remark 5.2. *Note that*

$$D = 2|\phi\rangle\langle\phi| - I = -\big(I - 2|\phi\rangle\langle\phi|\big).$$

Thus D differs from the reflection (defined in Lemma 4.1) $R_\phi := I - 2|\phi\rangle\langle\phi|$ by a global phase. Since global phase does not affect measurement probabilities, $G = DO_f$ has the same behaviour as $-R_\phi O_f$. Therefore analysis may be carried out using these reflections.

In Grover's algorithm, once the oracle has flipped the sign of the marked amplitude, the diffusion step reflects all amplitudes about the mean. This increases the marked amplitude and decreases the unmarked amplitudes. In this process, probability mass is concentrated on the marked state.

Grover's Algorithm: Then combining these reflections gives the Grover iterate,

$$G := DO_f = (2|\phi\rangle\langle\phi| - I)\, O_f.$$

Starting with $|\phi\rangle$, Grover's algorithm applies G repeatedly. Then measure in the computational basis. The k-th iterate is

$$|\psi_k\rangle = G^k|\phi\rangle$$

[9, p. 156];[5, p. 54]

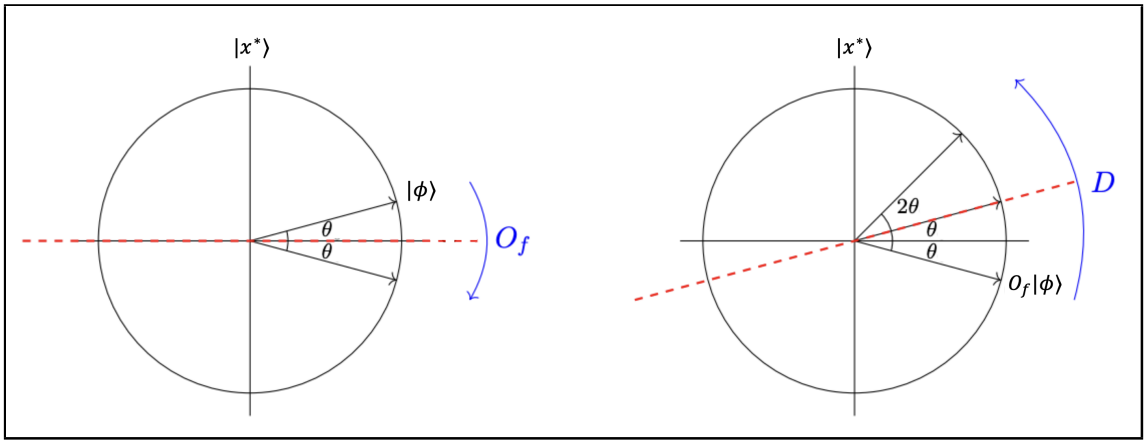


Figure 1: An illustration of the geometry of one Grover iterate in the two-dimensional subspace spanned by $|x^*\rangle$ and $|x^{*\perp}\rangle$. The oracle O_f reflects the initial state $|\phi\rangle$, and the diffusion operator D then reflects again so that $G = DO_f$ acts as a rotation by 2θ towards the marked state $|x^*\rangle$. Figure adapted from [8, p. 39].

Lemma 5.1 (The diffusion operator fixes $|\phi\rangle$ and reflects its orthogonal complement)**.** *Let*

$$D = 2|\phi\rangle\langle\phi| - I.$$

Then:

1. $D|\phi\rangle = |\phi\rangle$.
2. *If* $\langle\phi \mid \psi\rangle = 0$, *then* $D|\psi\rangle = -|\psi\rangle$.

[5, p. 69]

Proof. This follows from a standard calculation.

Since $\langle \phi \mid \phi \rangle = 1$,

$$D|\phi\rangle = (2|\phi\rangle\langle\phi| - I)|\phi\rangle = 2|\phi\rangle\langle\phi|\phi\rangle - |\phi\rangle = 2|\phi\rangle - |\phi\rangle = |\phi\rangle.$$

And if $\langle \phi \mid \psi \rangle = 0$, then

$$D|\psi\rangle = (2|\phi\rangle\langle\phi| - I)|\psi\rangle = 2|\phi\rangle\langle\phi|\psi\rangle - |\psi\rangle = -|\psi\rangle.$$

□

Lemma 5.2 (Two-dimensional invariant subspace)**.** *Let*

$$a := \langle x^* \mid \phi \rangle = \frac{1}{\sqrt{N}}, \qquad |x^{*\perp}\rangle := \frac{|\phi\rangle - a|x^*\rangle}{\sqrt{1-a^2}},$$

where $|x^{*\perp}\rangle$ is a unit vector orthogonal to $|x^*\rangle$. Then the subspace

$$\mathrm{span}\{|x^*\rangle, |x^{*\perp}\rangle\}$$

is invariant under the Grover iterate $G = DO_f$. [5, p. 54]; [8, p. 39]

Proof. $|x^*\rangle$ and $|x^{*\perp}\rangle$ form an orthonormal basis for the two-dimensional subspace, and

$$|\phi\rangle = a|x^*\rangle + \sqrt{1-a^2}\,|x^{*\perp}\rangle,$$

so $|\phi\rangle \in \mathrm{span}\{|x^*\rangle, |x^{*\perp}\rangle\}$. Hence

$$\mathrm{span}\{|x^*\rangle, |x^{*\perp}\rangle\} = \mathrm{span}\{|x^*\rangle, |\phi\rangle\}.$$

First consider O_f. Since there is a unique marked item,

$$O_f|x^*\rangle = -|x^*\rangle.$$

O_f also fixes every computational basis state orthogonal to $|x^*\rangle$, so

$$O_f|x^{*\perp}\rangle = |x^{*\perp}\rangle.$$

Thus O_f preserves $\mathrm{span}\{|x^*\rangle, |x^{*\perp}\rangle\}$.

Now consider $D = 2|\phi\rangle\langle\phi| - I$. Since $|\phi\rangle \in \mathrm{span}\{|x^*\rangle, |x^{*\perp}\rangle\}$, the space spanned by $|\phi\rangle$ lies in $\mathrm{span}\{|x^*\rangle, |x^{*\perp}\rangle\}$. D acts as the identity on $|\phi\rangle$ and as $-I$ on vectors orthogonal to $|\phi\rangle$. Therefore D preserves the plane $\mathrm{span}\{|x^*\rangle, |x^{*\perp}\rangle\}$.

Since both O_f and D preserve $\mathrm{span}\{|x^*\rangle, |x^{*\perp}\rangle\}$, so does their product $G = DO_f$. Thus $\mathrm{span}\{|x^*\rangle, |x^{*\perp}\rangle\}$ is invariant under G. [8, p. 39] □

The vectors $|x^*\rangle$ and $\left|x^{*\perp}\right\rangle$ form an orthonormal basis for the two-dimensional invariant subspace $\text{span}\{|x^*\rangle, |x^{*\perp}\rangle\}$. Since

$$\langle x^* \mid \phi\rangle = \frac{1}{\sqrt{N}},$$

it makes sense to define an angle $\theta \in (0, \pi/2)$ by

$$\sin\theta = \frac{1}{\sqrt{N}}.$$

Then

$$|\phi\rangle = \sin\theta\,|x^*\rangle + \cos\theta\left|x^{*\perp}\right\rangle.$$

So the initial state lies in the spanned plane, and both the oracle and diffusion (and any composition of the two) preserve the plane. Therefore, the Grover iterate acts as a rotation in the two-dimensional subspace.

5.3 Success probability and query complexity

Lemma 5.3 (Success probability after k Grover iterations)**.** *Let $N = 2^n$, and suppose there is one marked state $|x^*\rangle$. Define the angle by*

$$\sin\theta = \sqrt{1/N}$$

After applying the Grover iterate k times to the initial state,

$$|\phi\rangle := \frac{1}{\sqrt{N}} \sum_{x\in\{0,1\}^n} |x\rangle,$$

up to global phase, the state becomes

$$|\psi_k\rangle = \sin((2k+1)\theta)|x^*\rangle + \cos((2k+1)\theta)|x^{*\perp}\rangle,$$

where $\left|x^{*\perp}\right\rangle := \frac{|\phi\rangle - \langle x^*|\phi\rangle|x^*\rangle}{\sqrt{1-|\langle x^*|\phi\rangle|^2}}$.

Hence the success probability is

$$P_k = \mathbb{P}(\textit{outputting } x^*) = \sin^2((2k+1)\theta).$$

[5, p. 55-56]

Proof. By Lemma 5.2, the subspace $\text{span}\{|x^*\rangle, |x^{*\perp}\rangle\}$ is invariant under the Grover iterate G. It follows from

$$\langle x^*|\phi\rangle = \frac{1}{\sqrt{N}} \sum_x \delta_{x,x^*} = \frac{1}{\sqrt{N}} = \sin\theta,$$

that $|\phi\rangle = \sin\theta\,|x^*\rangle + \cos\theta\,\left|x^{*\perp}\right\rangle$.

Since

$$D = -(I - 2|\phi\rangle\langle\phi|),$$

the diffusion operator differs from the reflection $R_\phi := I - 2|\phi\rangle\langle\phi|$ by a global phase. Hence, up to global phase, the Grover iterate acts as the product of two reflections, and therefore by Theorem 4.1, acts as a rotation by angle 2θ on the two-dimensional invariant subspace. After k applications, the total rotation angle is $\theta + 2k\theta = \theta(2k+1)$. Hence, up to global phase,

$$G^k\,|\phi\rangle = \sin((2k+1)\theta)|x^*\rangle + \cos((2k+1)\theta)|x^{*\perp}\rangle.$$

[2, p.7] Finally, the success probability is the squared absolute value of the amplitude of $|x^*\rangle$. So it follows that

$$P_k = |\langle x^*|\psi_k\rangle|^2 = \sin^2((2k+1)\theta).$$

[5, p. 55-56] □

Remark 5.3. *Lemma 5.3 can be easily extended to a problem with $M > 1$ marked states by replacing $\sin\theta = \sqrt{1/N}$ with $\sin\theta = \sqrt{M/N}$.*

Example 5.1 (N=4)**.** *Consider the case where $N = 4$. Then the basis states are*

$$|00\rangle, |01\rangle, |10\rangle \text{ and } |11\rangle.$$

Suppose the marked state is $|x^*\rangle = |00\rangle$. The initial superposition is the state

$$|\phi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle),$$

so that each state begin with an amplitude $\frac{1}{2}$.

After applying the oracle, only the sign of the marked state flips. The state then becomes

$$\frac{1}{2}(-|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

Thus the amplitudes are $(-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ which average $\frac{1}{4}$.

The diffusion step reflects each amplitude about this average. Reflecting $-\frac{1}{2}$ and $\frac{1}{2}$ about $\frac{1}{4}$ gives 1 and 0 respectively. So the amplitudes are $(1, 0, 0, 0)$, and the state is

$$|00\rangle.$$

Therefore when $N = 4$, a single Grover iteration can find the marked state with probability 1.

5.4 Optimality of Grover's algorithm

Corollary 5.1 (The optimal number of Grover iterations k^*)**.** *The success probability after k Grover iterations is*

$$p_k = \sin^2((2k+1)\theta), \qquad \textit{where } \sin\theta = \sqrt{1/N}.$$

This is maximised when k is the integer closest to $\frac{\pi}{4\theta} - \frac{1}{2}$.

For large N, since $\theta \sim \sqrt{1/N}$, this gives

$$k^* \approx \frac{\pi}{4}\sqrt{N}$$

[2, p.118]

Remark 5.4. *The probability of success does not continue to increase forever. Due to the nature of the sine function, the probability oscillates as k increases. Thus, iterating past the optimal point causes rotation away from the marked point. This is called overshooting.*

Proof. $p_k = \sin^2((2k+1)\theta)$ is maximised when $(2k+1)\theta \approx \frac{\pi}{2}$. This gives $k \approx \frac{\pi}{4\theta} - \frac{1}{2}$.

In our unique marked case, $\sin\theta = \frac{1}{\sqrt{N}}$, so for large N it follows that $\theta \approx \frac{1}{\sqrt{N}}$. Therefore,

$$k = \mathcal{O}(\sqrt{N}),$$

and this yields the approximation $k^* \approx \frac{\pi}{4}\sqrt{N}$. [10, p. 30] □

In the black box model, the notion of quadratic improvement is only meaningful when compared under the same access assumptions. In classical computing, a unstructured search problem with a unique marked item may require you to search through all N items to find the marked item in the worst case. Any classical randomized algorithm would need $\Omega(N)$ queries to find the unique marked item with constant success probability.

By contrast, Grover's algorithm uses only $\mathcal{O}(\sqrt{N})$ oracle queries to reach a constant success probability. Thus, Grover speeds up unstructured search by reducing the number of queries needed from $\mathcal{O}(N)$ to $\mathcal{O}(\sqrt{N})$ which gives a quadratic improvement.

Theorem 5.1 (Grover lower bound)**.** *Any black box quantum algorithm that finds the unique marked item x^* with constant success probability must make at least $\Omega(\sqrt{N})$ oracle queries. [9, p. 189-190]*

Sketch proof. For an item x, let $|\phi_T^x\rangle$ denote the final state of the quantum algorithm after T queries to the oracle. For the algorithm to identify the marked item with constant success probability, the states $|\phi_T^x\rangle$ must be distinguishable for different values of x.

The idea is that one oracle query can change the state by only a limited amount because it flips the phase of only one basis state. Unless significant amplitude has been concentrated on that state, the effect of the query will be small. Averaging over all the possible marked items, one finds that each query increases distinguishability by at most $O(\sqrt{1/N})$.

Hence after T queries the total separation between the final states is at most

$$O(T\sqrt{1/N})$$

For the algorithm to work and identify the marked item with constant success probability, the final states which correspond to different marked items must differ by a constant (distinguishable) amount. Therefore it is required that

$$\mathcal{O}\left(\frac{T}{\sqrt{N}}\right) \geq c$$

for some constant $c > 0$, which implies that

$$T = \Omega(\sqrt{N})$$

[9, p. 190-191] □

In conclusion, Grover's algorithm achieves the optimal query complexity for unstructured search. This gives a quadratic improvement compared to the classical black box setting.

6 Original component: numerical study of amplitude amplification

Sections 4 and 5 are purely theoretical; I show that Grover's algorithm works with the state remaining in a two-dimensional invariant subspace spanned by the marked state and a state orthogonal to it, and rotates by angle 2θ under each Grover iterate. We found that the success probability after k iterations is $\sin^2((2k+1)\theta)$; therefore it is expected to oscillate. In this section I give an original study of these mechanisms.

The aim is not to reprove any theory, but to demonstrate and test the predictions. First, that repeated Grover iterates increase the probability of success by rotating towards the marked space. Second, that iterating past the optimum causes overshooting. These simulations provide quantitative evidence and intuition for amplitude amplification.

These simulations and figures have been coded by myself in python and full code is shown in the appendix.

6.1 Experimental design

I consider the unique marked version of Grover search with unique marked item x^*, and study the algorithm's behaviour for a range of values of N. For each N my code uses the formulas derived in section 5:

$$P_k = \sin^2((2k+1)\theta), \quad \sin\theta = \frac{1}{\sqrt{N}}.$$

Thus, this numerical study is not a separate model, but a direct implementation of section 5.

The code produces two plots and is shown in full in Appendix B.

First, for a fixed N, I compute the probability P_k for each $k = 0, 1, ..., k_{max}$, so that k_{max} is chosen to slightly extend past the optimum. This allows the decline in success probability to be seen. This decline is called overshooting and comes from when the algorithm has iterated past its optimum.

Second, I represent the two-dimensional subspace spanned by the marked state $|x^*\rangle$ and its orthogonal complement $|x^{*\perp}\rangle$. After k Grover iterations, the state has coordinates

$$x_k = \sin((2k+1)\theta), \quad y_k = \cos((2k+1)\theta),$$

which are the amplitudes of $|x^*\rangle$ and $|x^{*\perp}\rangle$ respectively. Plotting these coordinates provides a geometric visualization of the rotation under each Grover iterate.

For both figures, the code also computes the theoretically optimal number of Grover iterations,

$$k_*^{approx} = \frac{\pi}{4\theta} - \frac{1}{2},$$

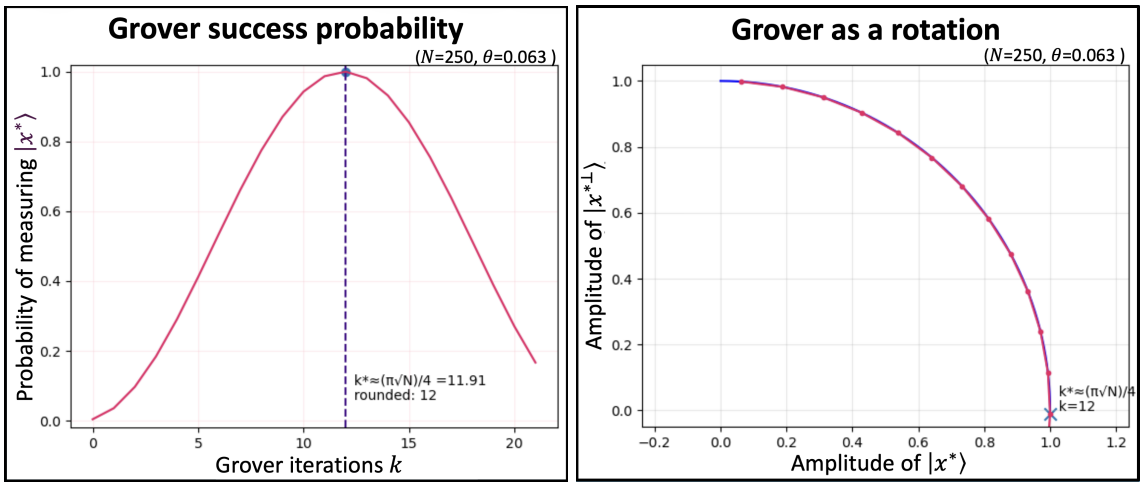
along with the closest integer k_*. This value is indicated by a vertical dotted line for the first plot and an x for the second. Since the theory predicts that the optimal is at

$$k_* \sim \frac{\pi}{4}\sqrt{N} \qquad \text{as } N \to \infty,$$

agreement between the plots and the predicted optimum provides numerical support for the $\sqrt{N}$ scaling.

6.2 Results

Here, N denotes the size of the database. Implementation using n qubits requires $2^n \geq N$.



(a) Peak near the predicted optimal k

(b) Geometric rotation in the subspace

Figure 2: Numerical illustrations of Grover's algorithm for $N = 250$.

Plot (a) shows the success probability is maximised near $k = 12$, which matches the theoretical optimum.
Plot (b) illustrates the geometric rotation picture from section 5: the two-dimensional subspace span$\{|x^*\rangle, |x^{*\perp}\rangle\}$. Each Grover iterate acts as a rotation by angle 2θ towards the marked item $|x^*\rangle$. Iterating past the optimum explains the overshooting seen in plot (a).

The rotation model depends only on N and remains valid as a two-dimensional amplitude amplification calculation, despite a literal n-qubit having $N = 2^n$ basis states.

7 Second mechanism: Quantum Phase Estimation

Grover's algorithm is an example of speedup through amplitude amplification. A second, and conceptually different mechanism is quantum phase estimation. In Grover search, the advantage comes from repeatedly rotating towards a marked state. By contrast, phase estimation gains information by combining unitary evolution with interference to extract useful information.

The point of this section is to not only present another algorithm, but to show that speedup can also come from spectral structure and the extraction of a property of a transformation, rather than from amplifying the probability of a desired basis state. This section therefore expands on the report's central theme by showing that quantum speedup is not tied to a single mechanism, but can come from fundamentally different mathematical structures.

7.1 Spectral Preliminaries

An essential structural fact for quantum phase estimation is that a unitary operator on a finite dimensional Hilbert space admits an orthonormal basis of eigenvectors, and that its

eigenvalues all lie on the unit circle. So, if U is a unitary, then there exists an orthonormal basis $\{|u_i\rangle\}$ along with real numbers $\phi_i \in [0, 1)$ such that

$$U|u_i\rangle = e^{2\pi i\phi_i}|u_i\rangle$$

for each i. In other words, applying U to an eigenvector is just multiplication by a phase. This is the form required for quantum phase estimation. It aims to extract the phase ϕ_i associated with the eigenvalue of U. The following lemma is the starting point of quantum phase estimation.

Lemma 7.1 (Eigen decomposition of a unitary)**.** *Let U be a unitary operator on a finite dimensional Hilbert space. Since U is normal, it is diagonalizable with respect to some orthonormal basis. Hence, U has an orthonormal basis of eigenvectors, and every eigenvalue λ of U satisfies $|\lambda| = 1$.*

Hence, every eigenvalue can be written in the form

$$\lambda = e^{2\pi i\phi}$$

for some $\phi \in [0, 1)$. [12, p. 72]

Proof. This is a consequence of the spectral theorem for Unitary operators [6, p. 3-4].

Notice that U is normal because U is a unitary, meaning $U^\dagger U = UU^\dagger = I$. Since U is normal, by the spectral theorem U is diagonalizable. Therefore there exists an orthonormal basis of eigenvectors of U.

If $|u\rangle$ is an eigenvector with eigenvalue λ, then

$$U|u\rangle = \lambda|u\rangle.$$

Since U is a unitary, it preserves norms. Hence,

$$\||u\rangle\| = \|U|u\rangle\| = \|\lambda|u\rangle\| = |\lambda|\||u\rangle\|.$$

Since $|u\rangle \neq 0$, we must have $|\lambda| = 1$. Any complex number of modulus 1 can be written as $e^{2\pi i\phi}$ for some $\phi \in [0, 1)$. □

7.2 Phase kickback and the controlled-U operator

In quantum phase estimation, the qubits are divided into two registers. The first register is *control register*, and its qubits determine which controlled powers of U are applied. The second is the *target register*, and it is the state that U acts on.

Remark 7.1. *For a controlled-U operation [12, p. 23], if the control qubit is in state $|0\rangle$,*

then the target register is left unchanged. However, if the control qubit is in state $|1\rangle$*, then* U *is applied to the target register. Thus, on basis states,*

$$|0\rangle \otimes |\psi\rangle \mapsto |0\rangle \otimes |\psi\rangle, \quad |1\rangle \otimes |\psi\rangle \mapsto |1\rangle \otimes U|\psi\rangle.$$

By linearity, this determines what the controlled-U *operator does to superpositions. In particular, when the control bit is in a superposition of the basis states and the target register is an eigenstate of* U*, the phase of the eigenvalue can be transferred to the control bit as a relative phase. This is known as* phase kickback*.*

Lemma 7.2 (Phase kickback)**.** *Let* U *be a unitary operator, and let* $|u\rangle$ *be an eigenvector of* U *with eigenvalue* $e^{2\pi i\phi}$*, so that*

$$U|u\rangle = e^{2\pi i\phi}|u\rangle \quad \textit{for some } \phi \in [0, 1).$$

Then applying a controlled-U *operation to the state*

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |u\rangle,$$

where the first qubit is the control qubit, yields

$$\frac{|0\rangle + e^{2\pi i\phi}|1\rangle}{\sqrt{2}} \otimes |u\rangle.$$

Thus the eigenvalue phase is a relative phase on the control qubit. [13, p. 3] [4, p. 35]

This shows that when the target register is prepared in an eigenstate of U, the application of U transfers the phase onto the $|1\rangle$ component of the control qubit. The phase is "kicked back" from the target register to the control register.

Proof. By linearity

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |u\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |u\rangle + |1\rangle \otimes |u\rangle).$$

A controlled-U operation leaves the target register unchanged when the control qubit is $|0\rangle$, but will apply U when the control qubit is $|1\rangle$. So,

$$U_c(\frac{1}{\sqrt{2}}(|0\rangle \otimes |u\rangle + |1\rangle \otimes |u\rangle)) = \frac{1}{\sqrt{2}}(|0\rangle \otimes |u\rangle + |1\rangle \otimes U|u\rangle),$$

where U_c is the controlled-U operator.

Since $|u\rangle$ is an eigenvector with eigenvalue $e^{2\pi i\phi}$, $U|u\rangle = e^{2\pi i\phi}|u\rangle$.

Substituting this gives

$$\frac{1}{\sqrt{2}}\Big(|0\rangle \otimes |u\rangle + e^{2\pi i\phi}|1\rangle \otimes |u\rangle\Big) = \frac{|0\rangle + e^{2\pi i\phi}|1\rangle}{\sqrt{2}} \otimes |u\rangle$$

[14, p. 1-2] □

7.3 Quantum Phase Estimation algorithm

The structure of the quantum phase estimation algorithm is now described. Let U be a unitary operator and suppose $|u\rangle$ is an eigenvector of U with eigenvalue $e^{2\pi i\phi}$, where $\phi \in [0, 1)$. The quantum phase estimation algorithm estimates the phase ϕ in the equation

$$U|u\rangle = e^{2\pi i\phi}|u\rangle, \qquad \phi \in [0, 1).$$

The algorithm uses two registers:

- a control register of m qubits, initially in the state $|0\rangle^{\otimes m}$,
- a target register in the eigenstate $|u\rangle$.

[12, p. 221]

Then, the initial joint state is

$$|0\rangle^{\otimes m} \otimes |u\rangle.$$

Following Vazirani [14, p. 2–3], the quantum phase estimation algorithm has four steps.

Step 1: Create a uniform superposition. Applying a Hadamard gate (recall Example 4.1) to each qubit in the control register. This produces the superposition

$$\frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} |k\rangle \otimes |u\rangle.$$

The state is now in an equal superposition of all computational basis states.

Step 2: Encode the phase. The next step is to apply the controlled powers of U. More precisely, we apply a controlled-U^{2^j} operation to the target register controlled by the j-th control qubit. Since $|u\rangle$ is an eigenvector of U, it is also an eigenvector of U^{2^j}, with eigenvalue $e^{2\pi i 2^j\phi}$ so that

$$U^{2^j}|u\rangle = e^{2\pi i 2^j\phi}|u\rangle.$$

By phase kickback, every controlled operation contributes its corresponding phase to the control register. The phase is encoded in the amplitudes of the control register, while the

target register remains in state $|u\rangle$. After all the controlled powers have been applied, the joint state is

$$\frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} e^{2\pi i k\phi} |k\rangle \otimes |u\rangle.$$

The control register is now in a phase-encoded superposition. This is called the Fourier state since it has the same form that appears in the quantum Fourier transform.

Step 3: Apply the inverse quantum Fourier transform. At this stage the phase information is encoded in the amplitudes of the control register. Applying the inverse quantum Fourier transform to the control register (QFT^{-1}) converts this phase-encoded state into one where measurement in the computational bits yields an approximation of ϕ.

Step 4: Measure. Finally, we measure the control register in the computational basis. The outcome gives us an m-bit approximation of ϕ. If ϕ can be written in exactly m bits, then the algorithm returns exactly ϕ. Otherwise, the measurement outcome is, with high probability, an m-bit approximation of ϕ.

7.4 Accuracy and Success probability

The previous subsection showed that before measurement, the joint state of the control and target registers is

$$\frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} e^{2\pi i k\phi} |k\rangle \otimes |u\rangle.$$

After applying the inverse quantum Fourier transform, we measure the control register in the computational basis. In this section we look at how likely this measurement is to return a good approximation to the phase ϕ.

Theorem 7.1. *Let n be the closest integer to $2^m\phi$. Then after applying the inverse quantum Fourier transform, the probability that measuring the control register yields n is at least $4/\pi^2$. [11, p. 29]*

Proof sketch. After applying QFT^{-1}, the amplitude of outcome y can be written as a finite geometric sum. For $y = n$, this sum can be written explicitly and bounded below using inequalities. This gives

$$\mathbb{P}(n) \geq \frac{4}{\pi^2}.$$

Due to the length, a full proof is given in Appendix C. [11, p. 29] □

7.5 How QPE is a different speedup mechanism

Quantum phase estimation illustrates a different mechanism of quantum speedup from the one seen in Grover's algorithm. In Grover search, the advantage comes from amplitude amplification, where the algorithm repeatedly applies the Grover iterate so that the amplitude of the marked state increases, making the measurement more likely to return the desired outcome. This is done in the black box query model, where we are interested in how many oracle calls are needed to identify the marked item.

By contrast, quantum phase estimation does not work by amplifying the probability of a state. Instead, it uses controlled powers of a unitary operator and the inverse quantum Fourier transform to extract spectral information (eigenphase). So phase estimation uses spectral structure instead of amplitude amplification.

This contrast is significant because it shows that quantum speedup is not constrained to a single mechanism. Grover's algorithm is advantageous by reshaping probability distributions for measurement, while phase estimation is a way to extract global information about a transformation from unitaries. For this reason, phase estimation is useful in more structured quantum algorithms, where the task is to investigate a property of the operator rather than to search through objects one by one. Therefore this section supports the central theme of this report: quantum speedup can come from fundamentally different mathematical mechanisms.

8 Conclusions

8.1 Limits of quantum algorithmic speedup

Quantum speedup is only meaningful once a cost model has been fixed. In this report, two have been used: (i) the black-box query model, where the cost is the oracle calls, and (ii) circuit-level resources like gate count/length as a secondary model. Within the query model, unstructured search illustrates a clear limit: although amplitude amplification achieves a quadratic improvement, the lower bound results show that there is no quantum algorithm that can solve unique unstructured search with constant success probability using $o(\sqrt{N})$ oracle queries. Given this, the $\Theta(\sqrt{N})$ scaling is optimal for this task.

More, generally, quantum speedups are constrained by measurement. Although an n-qubit state has 2^n complex amplitudes, a measurement reveals only limited classical information per run. This is why quantum algorithms must typically either reshape the probability mass so that the good outcome is observed with high probability (amplitude amplification), or use a global property to get a small number of measurable bits (phase estimation extracts eigenphase information). These constraints help explain why many quantum algorithms need some algebraic or spectral structure, and why only limited improvements can be made on unstructured problems.

8.2 Summary

This report compared two fundamentally different avenues to quantum speedup under matched model assumptions. In unstructured search, repeated application of the Grover iterate (a product of two reflections) causes a rotation in a two-dimensional invariant subspace, giving a success probability $P_k = \sin^2((2k+1)\theta)$ and optimal oracle query complexity $\Theta(\sqrt{N})$. The simulation reproduced the peak near $\frac{\pi}{4\theta} - \frac{1}{2}$ and illustrated overshooting when iterating too far. For phase estimation, speedup instead comes from controlled powers of a unitary and interference via the inverse QFT, allowing the extraction of spectral information with a constant lower bound success for the nearest m-bit estimate. Together these examples support the report in that quantum speedup does not arise from a single phenomenon but could arise from multiple distinct mechanisms. A natural extension would be to broaden the numerical study to the case with multiple marked items.

A Full proof of Theorem 4.1

Proof of Theorem 4.1. R_u sends u to $-u$ but fixes every vector orthogonal to u, and similarly for R_v. Hence any vector $w \in \text{span}\{|u\rangle, |v\rangle\}^\perp$ is orthogonal to both u and v, so we have

$$R_u w = w \text{ and } R_v w = w$$

Therefore $(R_u R_v)w = w$, proving $(R_u R_v)|_{\text{span}\{|u\rangle,|v\rangle\}^\perp} = I$.

Since $|u\rangle$ and $|v\rangle$ are linearly independent, the subspace

$$\text{span}\{|u\rangle, |v\rangle\}$$

is two-dimensional. Since R_v depends on only the projector $|v\rangle\langle v|$, replacing $|v\rangle$ by a global phase does not change R_v. Hence we may assume that $\langle u|v\rangle \in [0,1]$ is real, so

$$\langle u|v\rangle = \cos\theta.$$

Then it can be assumed that

$$\langle u|v\rangle = \cos\theta, \qquad \theta \in (0, \pi/2].$$

Define

$$|u_\perp\rangle := \frac{|v\rangle - \cos\theta\,|u\rangle}{\sin\theta}.$$

Then $\langle u|u_\perp\rangle = 0$ and $\|u_\perp\| = 1$ by

$$\langle u|u_\perp\rangle = \left\langle u \,\middle|\, \frac{v - \cos\theta u}{\sin\theta} \right\rangle = \frac{\langle u \mid v\rangle - \cos\theta\langle u \mid u\rangle}{\sin\theta} = \frac{\cos\theta - \cos\theta \cdot 1}{\sin\theta} = 0$$

and

$$\begin{aligned}\|u_\perp\|^2 &= \langle u_\perp \mid u_\perp\rangle \\ &= \frac{1}{\sin^2\theta}\langle v - \cos\theta\, u \mid v - \cos\theta\, u\rangle \\ &= \frac{1}{\sin^2\theta}\big(\langle v|v\rangle - \cos\theta\langle v|u\rangle - \cos\theta\langle u|v\rangle + \cos^2\theta\langle u|u\rangle\big) \\ &= \frac{1}{\sin^2\theta}\big(1 - \cos^2\theta - \cos^2\theta + \cos^2\theta\big) \\ &= \frac{1-\cos^2\theta}{\sin^2\theta} = 1.\end{aligned}$$

so $\{|u\rangle, |u_\perp\rangle\}$ is an orthonormal basis of $\operatorname{span}\{|u\rangle, |v\rangle\}$.

Step 1: compute $R_v|u\rangle$ and $R_v|u_\perp\rangle$.

$$R_v|u\rangle = |u\rangle - 2|v\rangle\langle v|u\rangle = |u\rangle - 2|v\rangle\cos\theta = |u\rangle - 2\cos\theta(\cos\theta|u\rangle + \sin\theta|u_\perp\rangle),$$

so

$$R_v|u\rangle = (1 - 2\cos^2\theta)|u\rangle - 2\sin\theta\cos\theta\,|u_\perp\rangle = -\cos(2\theta)|u\rangle - \sin(2\theta)|u_\perp\rangle.$$

And, using

$$R_v|u_\perp\rangle = |u_\perp\rangle - 2|v\rangle\langle v|u_\perp\rangle \text{ and } \langle v|u_\perp\rangle = \sin\theta$$

gives

$$R_v|u_\perp\rangle = |u_\perp\rangle - 2\sin\theta(\cos\theta|u\rangle + \sin\theta|u_\perp\rangle) = -\sin(2\theta)|u\rangle + \cos(2\theta)|u_\perp\rangle.$$

Step 2: apply R_u. Using

$$R_u|u\rangle = -|u\rangle, \qquad R_u|u_\perp\rangle = |u_\perp\rangle,$$

Gives

$$R_uR_v|u\rangle = R_u\big(-\cos(2\theta)|u\rangle - \sin(2\theta)|u_\perp\rangle\big) = \cos(2\theta)|u\rangle - \sin(2\theta)|u_\perp\rangle,$$

and

$$R_uR_v|u_\perp\rangle = R_u\big(-\sin(2\theta)|u\rangle + \cos(2\theta)|u_\perp\rangle\big) = \sin(2\theta)|u\rangle + \cos(2\theta)|u_\perp\rangle.$$

Thus, in the orthonormal basis $\{|u\rangle, |u_\perp\rangle\}$, the matrix of $(R_uR_v)\big|_{\operatorname{span}\{|u\rangle,|v\rangle\}}$ is

$$\begin{pmatrix}\cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta\end{pmatrix},$$

which is exactly the matrix of a rotation by angle 2θ. This completes the proof. [12, p. 252] □

B Section 6: original component full code

```python
import numpy as np
import matplotlib.pyplot as plt

def grover_success_probability(N: int, k_max: int = None):
    if N <= 1:
        raise ValueError("N must be at least 2")

    theta = np.arcsin(1 / np.sqrt(N))

    # show a little bit past the peak
    if k_max is None:
        k_max = int(np.ceil(1.25 * (np.pi / 4) * np.sqrt(N))) + 5

    k = np.arange(theta, k_max + 1)
    p = np.sin((2 * k + 1) * theta) ** 2

    # first peak iteration and closest integer
    k_star_cont = (np.pi / (4 * theta)) - 0.5
    k_star = int(np.round(k_star_cont))

    return k, p, theta, k_star_cont, k_star

def plot_grover(N: int, k_max: int = None):
    k, p, theta, k_star_cont, k_star = grover_success_probability(N, k_max)

    plt.figure()
    plt.plot(k, p, color = "#E91E63")
    plt.axvline(k_star, linestyle="--", color = "#4A148C")
    plt.scatter([k_star], [np.sin((2 * k_star + 1) * theta) ** 2])
    plt.xlabel("Grover iterations k")
    plt.ylabel("Probability of marked item sin²((2k+1)theta)")
    plt.title(f"Grover success probability for N={N}\n"
              f"sin(theta)=1/(N),  theta=(1/(N))={theta:.4g} rad")
    plt.text(k_star, 0.05,
             f"  k*(N)/4 ={k_star_cont:.2f}\n  rounded: {k_star}",
             va="bottom")
    plt.ylim(-0.02, 1.02)
    plt.grid(True, alpha=0.3, color = "pink")
    plt.show()
```

```python

def rotation_points(N: int, k_max: int = None):
    # If angle is |x*_perp> toward |x*> is (2k+1)theta, then:
    # x = sin((2k+1)theta)  = amplitude on |x*>
    # y = cos((2k+1)theta)  = amplitude on |x*_perp>
    k, _, theta, k_star_cont, k_star = grover_success_probability(N, k_max)
    ang = (2 * k + 1) * theta
    x = np.sin(ang)  # along |x*>
    y = np.cos(ang)  # along |x*_perp>
    return k, x, y, theta, k_star_cont, k_star

# unit circle and points
def plot_rotation(N: int, k_max: int = None):
    k, x, y, theta, k_star_cont, k_star = rotation_points(N, k_max)
    # do the first quadrant only
    t = np.linspace(theta, np.pi/2, 400)
    plt.figure()
    # curve of circle
    plt.plot(np.sin(t), np.cos(t), color = "blue")
    plt.plot(x, y, marker="o", markersize=3, color = "#E91E63")

    # mark the peak
    plt.scatter([x[k_star]], [y[k_star]], marker="x", s=80)
    plt.text(x[k_star], y[k_star],
             f"  k*(N)/4\n  k={k_star}",
             va="bottom")

    plt.xlabel("sin((2k+1)theta) (= amplitude of |x*⟩)")
    plt.ylabel("cos((2k+1)theta) (= amplitude of |x*⟩)")
    plt.title(f"Grover as a rotation in 2D plane with N={N}, theta={theta:.4g} rad)\n"
              f"Each iteration adds 2theta to the angle")
    plt.axis("equal")
    plt.xlim(-0.05, 1.05)
    plt.ylim(-0.05, 1.05)
    plt.grid(True, alpha=0.3)
    plt.show()

# Running the examples shown in the scholarly report:
# shows probability against iterations with k* marker
plot_grover(N=250)
# shows the rotation in the spanned plane
```

```
plot_rotation(N=250)
```

C Full proof of Theorem 7.1

Proof of theorem 7.1. Let $M = 2^m$. After the inverse quantum Fourier transform, the amplitude of the basis state $|y\rangle$ is

$$\alpha_y = \frac{1}{M} \sum_{k=0}^{M-1} e^{2\pi i k(\phi - y/M)}.$$

By the Born rule, the probability of measuring y is

$$\mathbb{P}[y] = |\alpha_y|^2.$$

Let n be the closest integer to $M\phi$, and write

$$M\phi = n + \varepsilon, \text{ where } |\varepsilon| \leq \tfrac{1}{2}$$

Then

$$\phi - \frac{n}{M} = \frac{\varepsilon}{M},$$

so

$$\alpha_n = \frac{1}{M} \sum_{k=0}^{M-1} e^{2\pi i k\varepsilon/M}.$$

If $\varepsilon = 0$, then $\alpha_n = 1$, so $\mathbb{P}[n] = 1$ so theres nothing to prove. So, assume that $\varepsilon \neq 0$.

Using the geometric series formula:

$$\sum_{k=0}^{M-1} z^k = \frac{1 - z^M}{1 - z},$$

with $z = e^{2\pi i\varepsilon/M}$, we get

$$\alpha_n = \frac{1}{M} \cdot \frac{1 - e^{2\pi i\varepsilon}}{1 - e^{2\pi i\varepsilon/M}}.$$

So,

$$\mathbb{P}[n] = \frac{1}{M^2} \frac{|1 - e^{2\pi i\varepsilon}|^2}{|1 - e^{2\pi i\varepsilon/M}|^2}.$$

And, using

$$|1 - e^{2\pi i x}| = 2|\sin(\pi x)|,$$

this then becomes

$$\mathbb{P}[n] = \frac{\sin^2(\pi\varepsilon)}{M^2 \sin^2(\pi\varepsilon/M)}.$$

Since $|\varepsilon| \leq \frac{1}{2}$, we have $0 \leq \pi|\varepsilon| \leq \pi/2$, and we get the inequalities

$$\frac{2}{\pi}\theta \leq \sin\theta \leq \theta \qquad (0 \leq \theta \leq \pi/2)$$

Using these bounds on the numerator and the denominator give

$$\sin(\pi|\varepsilon|) \geq 2|\varepsilon|, \qquad \sin(\pi|\varepsilon|/M) \leq \pi|\varepsilon|/M.$$

And finally we get

$$\mathbb{P}[n] \geq \frac{(2|\varepsilon|)^2}{M^2(\pi|\varepsilon|/M)^2} = \frac{4}{\pi^2}.$$

[4, p. 223];[11, p. 29] □

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